# HYP-1

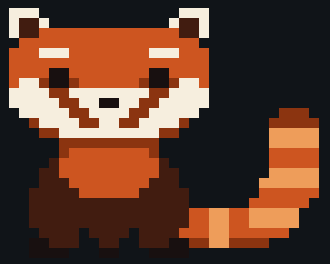
## FIELD MANUAL

VERSION 1.0.31

Hypnotic techno and acid firmware for the FM-1.
Every control numbered, every screen lettered.
Works the same on the device and in the web simulator.

rustypawz.github.io/fm1-HYP1 www.redtail.works youtube.com/@RedtailWorks

**Watch it played live:** youtube.com/watch?v=0UvGnZE2_VI



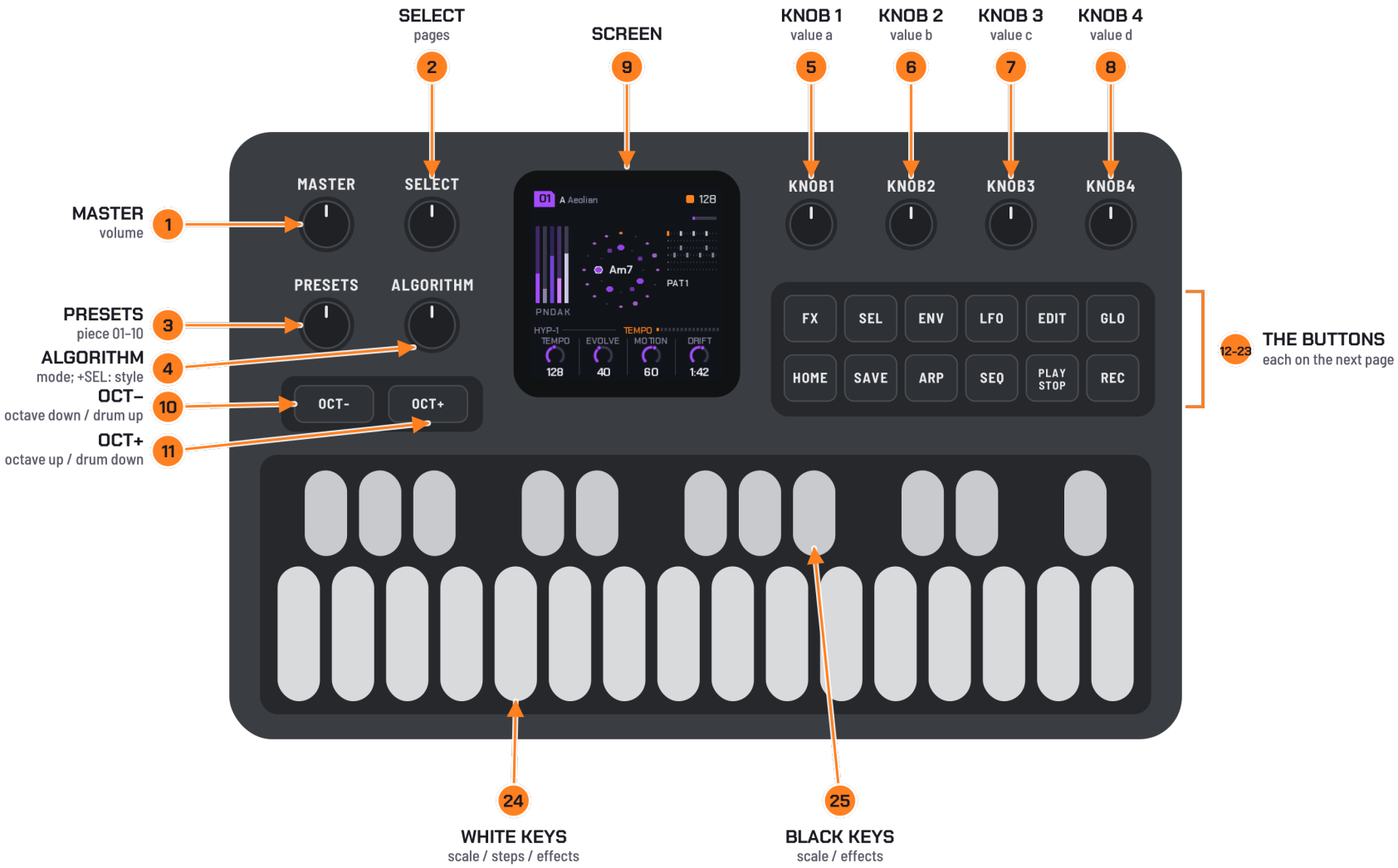
REDTAIL WORKS

PRESENTS

Numbers (1–25) are the device's controls. Letters (a, b, c…) are things on the screen.

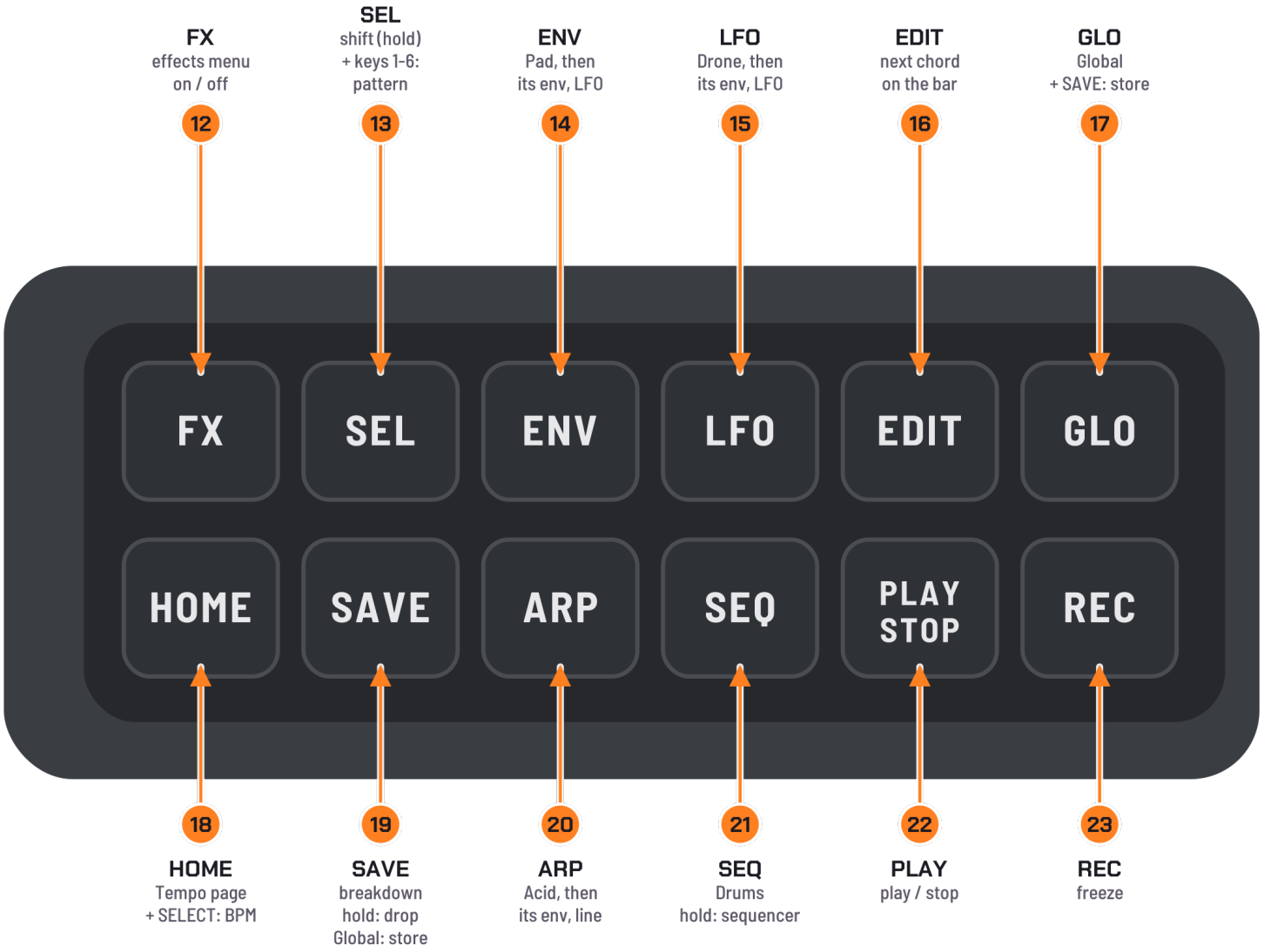
# The device

NUMBERS = CONTROLS · EACH ONE IS EXPLAINED ON THE NEXT PAGES



# The button panel, up close

ONE BUTTON CAN DO SEVERAL THINGS: TAP, TAP AGAIN, OR HOLD



# What each control does

THE NUMBERS FROM THE DEVICE PICTURES

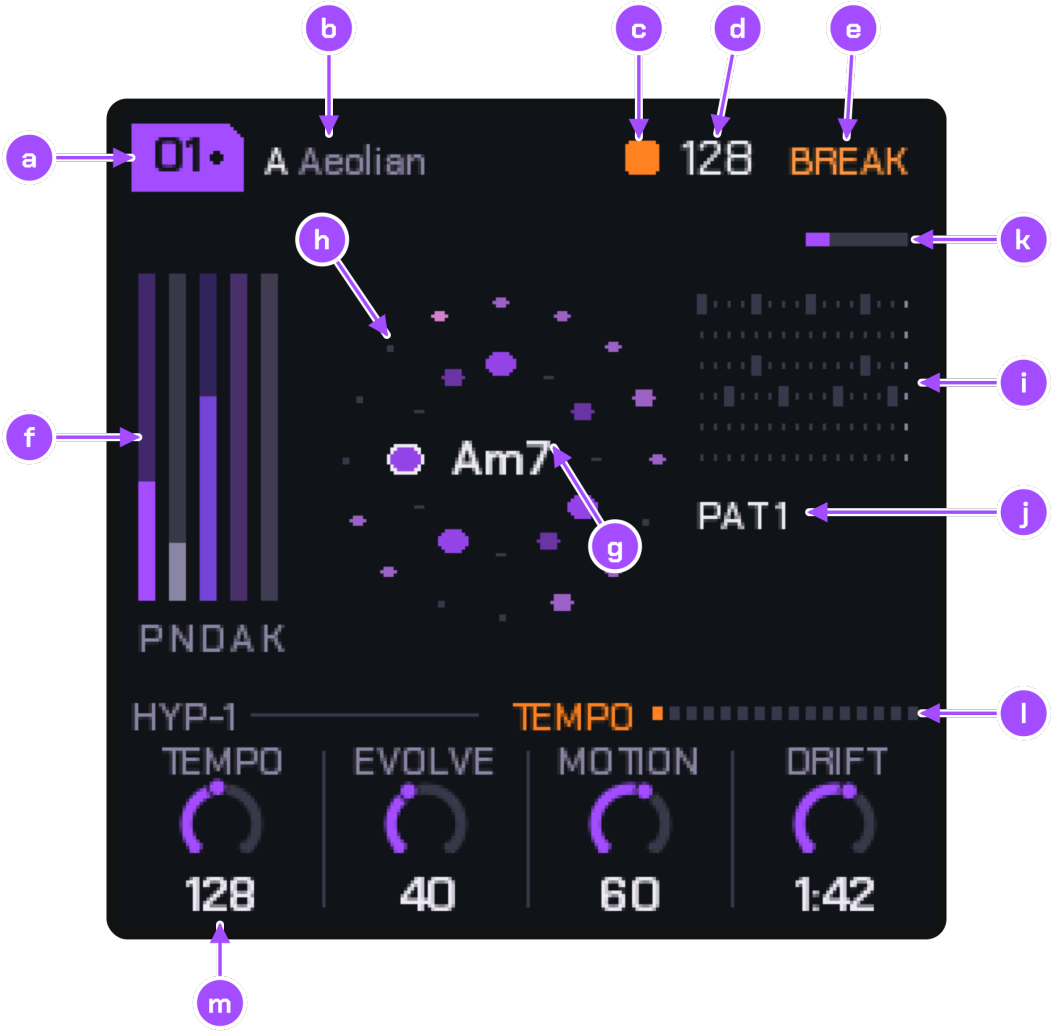
1. **MASTER.** Volume. Doesn't change the music, only how loud it is.
2. **SELECT.** Steps through every page in order (Tempo > Pad > … > Global). Hold HOME and turn it to set the tempo from any page.
3. **PRESETS.** Chooses one of the 10 pieces, 01–10. Each has its own settings, six drum patterns and drum sounds.
4. **ALGORITHM.** The mode of the scale: Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Pent major, Pent minor. With SEL held: the sound of the voice on screen (on the drums' pages, the whole kit).

5–8. **KNOB 1–4.** Change values a, b, c, d of whichever page is on screen (the four dials along the screen's bottom).

9. **Screen.** Shows the piece, key, tempo, section, the music moving, and the page you're on. See "The screen".
10. **OCT–.** An octave down (to –3): on a Pad, Drone or Acid page, that voice; on other pages, the whole song. Sequencer: the previous drum.
11. **OCT+.** An octave up (to +3), the same way. In the sequencer: the next drum.
12. **FX.** Turns the effects menu on and off. While it's on, the keys play effects.
13. **SEL.** Shift (hold). + white keys 1–6: pattern 1–6. + PRESETS: piece on the next bar. + ALGORITHM: the voice's sound. + ENV: the voice on screen by hand. + SEQ: links on/off. + GLO: remote (the panel plays the emulator over USB). Sequencer: + black 1–6 copy, + OCT– clear lane, + OCT+ restore.
14. **ENV.** Pad. Press again: Pad env. Again: Pad LFO. With SEL held, on any voice's page: that voice by hand (off the algorithm; again: back).
15. **LFO.** Drone. Press again: Drone env. Again: Drone LFO.
16. **EDIT.** Moves to the next chord on the next bar.
17. **GLO.** Global page: root, theme, tune, MIDI out. On this page SAVE stores the piece.
18. **HOME.** Tempo page. Hold it and turn SELECT to change the tempo from anywhere.
19. **SAVE.** Tap: breakdown from the next bar; tap again: 4-bar build, then the drop. Hold: the drop. In FX, held: lock mode (tap keys to lock). On Global: stores the piece (hold: factory).
20. **ARP.** Acid. Press again: Acid env. Again: Acid line.
21. **SEQ.** Drums, again Links, again Kit. Hold: the step sequencer. Tap while in the sequencer: back to Drums.
22. **PLAY/STOP.** Starts and stops.
23. **REC.** Freeze: holds everything on the chord it's on, until pressed again.
24. **White keys.** Normally: the notes of the scale (tap to add or remove; the root stays). On the pages of a voice played by hand: that voice. Sequencer: the 16 steps. FX: effects.
25. **Black keys.** Normally: notes of the scale; on the pages of a voice played by hand, that voice. FX: gates, throw, crush, fill, solos.

# The screen

LETTERS = WHAT'S ON THE SCREEN



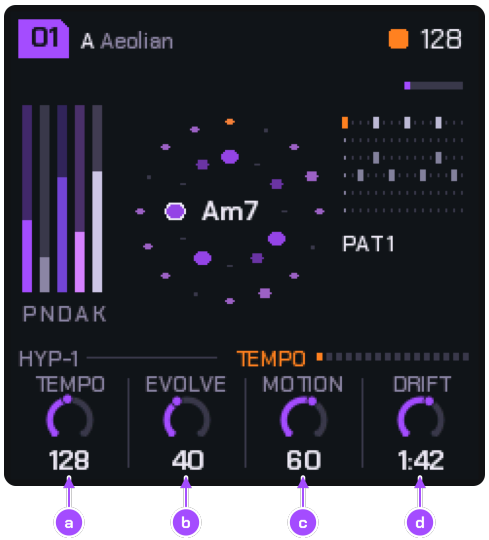

- **a** **Piece.** The piece playing, 01–10 (PRESETS). A dot in the tag: changes not yet stored.
- **b** **Key.** Root and mode (GLO > Root; ALGORITHM).
- **c** **Beat light.** Flashes on the beat.
- **d** **Tempo.** BPM. A small dot to its right: Theme, Tune or MIDI out waiting to be written (two seconds after a change).
- **e** **Section.** BREAK, BUILD, or blinking while it waits for the bar. Blank when the groove is running normally.
- **f** **Levels.** Meters for P pad, N noise, D drone, A acid, K kit (drums).
- **g** **Chord.** The chord now; the dots in the ring are the scale's notes.
- **h** **Acid loop.** The outer ring: one dot a step, bigger where the acid line plays.
- **i** **Drum pattern.** The six lanes of the pattern playing.
- **j** **Pattern number.** PAT1–6; shows the next one when a change is waiting for the bar.
- **k** **Drift bar.** How far through the current chord (to the next one).
- **l** **Page.** The page's name; one dot a page, the lit one is where you are. At the line's left: the sound of the voice on screen (drums: the kit).
- **m** **Knob cell.** Name, ring and value of what KNOB 1–4 changes. Every page has four: a, b, c, d.

# The pages

EACH PAGE HAS FOUR VALUES, a b c d — KNOBS 5, 6, 7, 8 CHANGE THEM

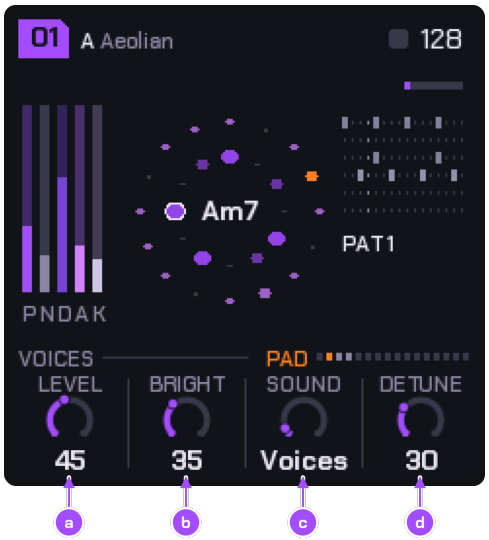
SELECT (2) steps through every page in this order. The buttons jump straight to a voice, and pressing the same button again steps through that voice's envelope and LFO. Changes stay with the piece until the power goes off. The sequencer, effects and themes follow.

## Tempo REACH IT: HOME




- **a** **Tempo.** 70–190 BPM. Also HOME + SELECT from any page.
- **b** **Evolve.** How restless the acid line and the chords are.
- **c** **Motion.** How far the slow tide swings between sparse and full.
- **d** **Drift.** How long a chord lasts, 20 s to 5 min. EDIT jumps to the next chord now.

## Pad REACH IT: ENV



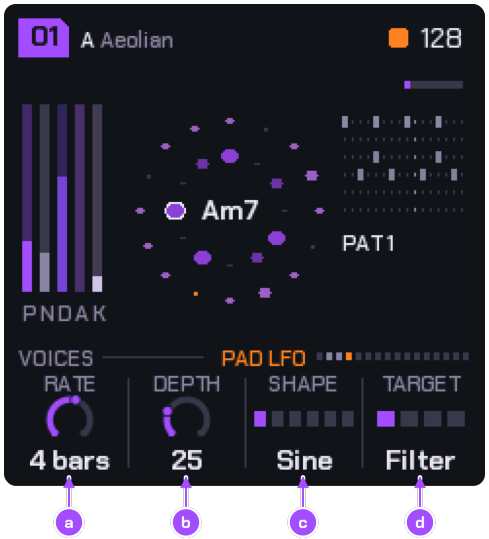

- **a** **Level.** The pad's volume.
- **b** **Bright.** Its filter and vowel.
- **c** **Sound.** Rhodes, Voices, Strings, Glass, Organ, Warm, Stabs, Dark.
- **d** **Detune.** Spreads the voices (and the strings) apart.

## Pad env REACH IT: ENV ×2



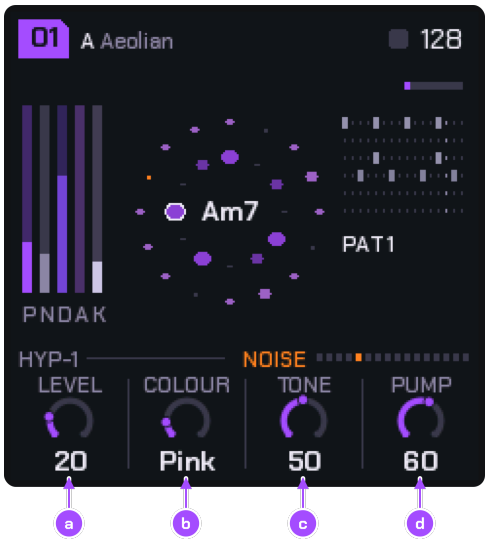

- **a** **Attack.** 0.05–10 s: how slowly each voice breathes in.
- **b** **Hold.** 0–8 s at full.
- **c** **Release.** 0.1–12 s: the fade out.
- **d** **Rest.** 0–6 s of silence before the next breath.

## Pad LFO REACH IT: ENV ×3



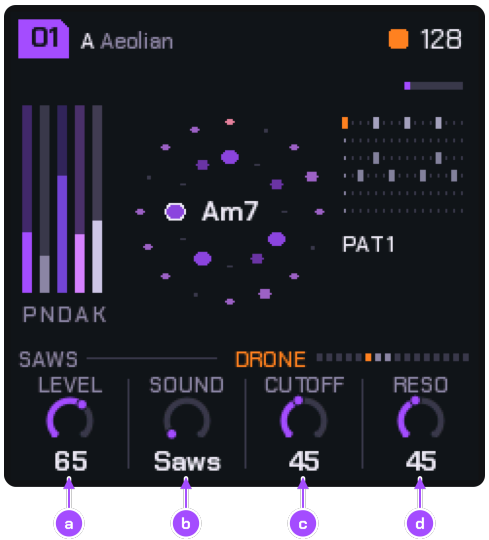

- **a** **Rate.** 1/4 note to 32 bars.
- **b** **Depth.** How much it moves.
- **c** **Shape.** Sine, Triangle, Saw, Square, S&H;, Drift.
- **d** **Target.** Filter, Level, Pan or Pitch.

## Noise REACH IT: SELECT



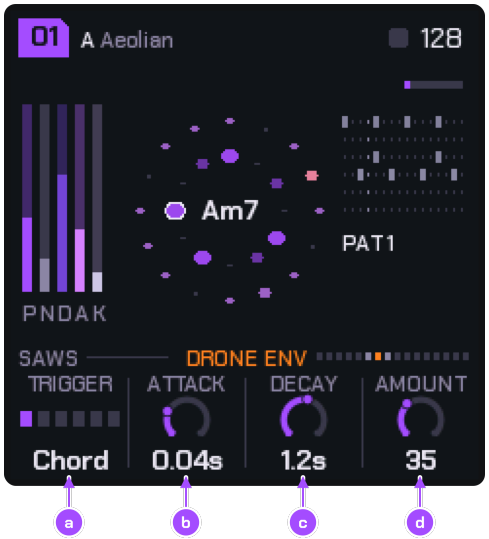

- **a** **Level.** A bed of grain under everything.
- **b** **Colour.** White, pink, brown, blue, violet, grey, black (crackle), yellow (radio hiss).
- **c** **Tone.** A low-pass on it, 300 Hz–19 kHz.
- **d** **Pump.** How hard the kick ducks it.

## Drone REACH IT: LFO



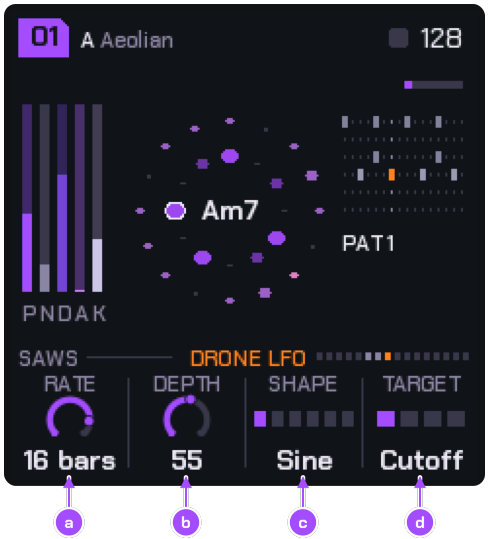

- **a** **Level.** The bass drone's volume.
- **b** **Sound.** Saws, Warm, Grit, Dirty, Reese, Pulse, FM, Organ. OCT–/+: its register.
- **c** **Cutoff.** Its filter.
- **d** **Reso.** The filter's resonance.

## Drone env REACH IT: LFO ×2



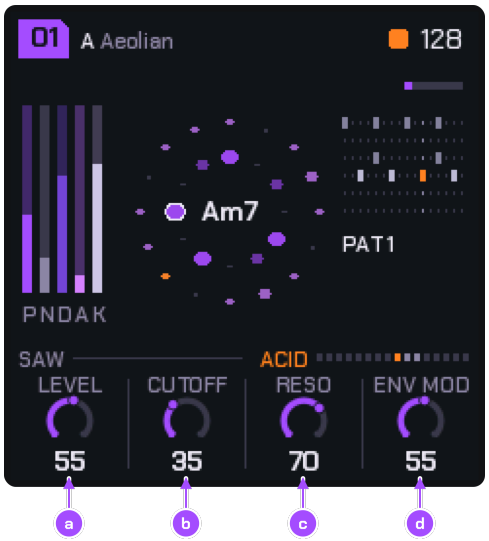

- **a** **Trigger.** Each chord, every 1, 2, 4 or 8 bars, or Off.
- **b** **Attack.** 0.01–4 s.
- **c** **Decay.** 0.05–16 s.
- **d** **Amount.** How far it opens the cutoff (up to 4 octaves).

## Drone LFO REACH IT: LFO ×3



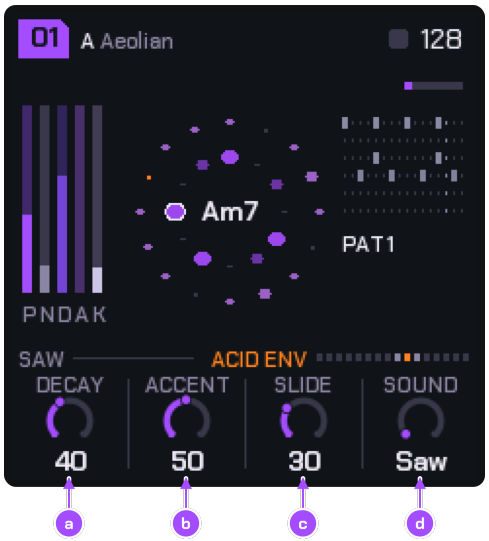

- **a** **Rate.** 1/4 note to 32 bars.
- **b** **Depth.** How much it moves.
- **c** **Shape.** Sine, Triangle, Saw, Square, S&H;, Drift.
- **d** **Target.** Cutoff, Pitch, Level or Drive. Default: a 16-bar sine on the cutoff.

## Acid REACH IT: ARP



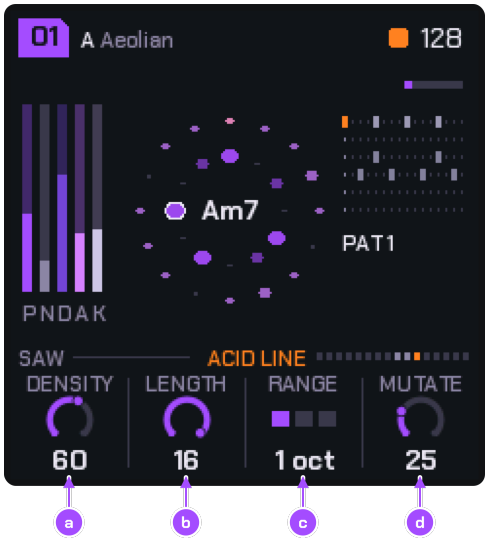

- **a** **Level.** The acid line's volume.
- **b** **Cutoff.** The filter.
- **c** **Reso.** Squelch.
- **d** **Env mod.** How far each note opens the filter.

## Acid env REACH IT: ARP ×2



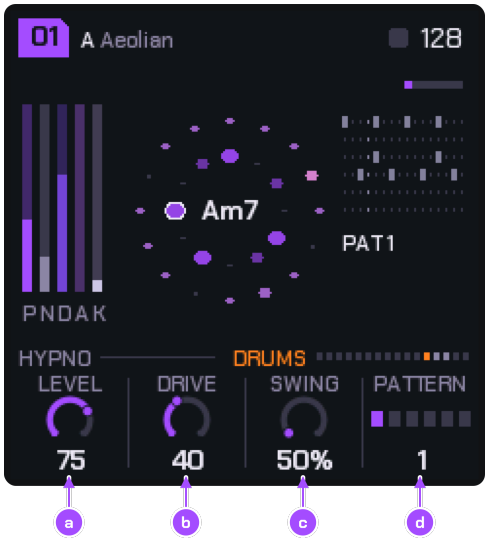

- **a** **Decay.** 0.06–1.9 s.
- **b** **Accent.** How hard accented steps hit.
- **c** **Slide.** Glide between tied notes.
- **d** **Sound.** Saw, Square, Driven, Sync, Pluck, PWM, Chip, Sub.

## Acid line REACH IT: ARP ×3



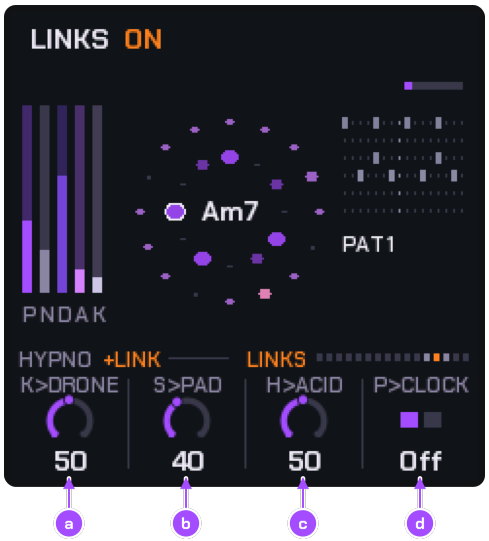

- **a** **Density.** How many steps play.
- **b** **Length.** 1–16 steps; odd lengths turn against the bar.
- **c** **Range.** 1–3 octaves.
- **d** **Mutate.** How often steps are made anew. OCT−/+ move it an octave.

## Drums REACH IT: SEQ




- **a** **Level.** The kit's volume.
- **b** **Drive.** The kick's drive.
- **c** **Swing.** 50–75%: the off-sixteenths late.
- **d** **Pattern.** 1–6. Waits for the end of the bar, so turning past others plays none of them.

## Links REACH IT: SEQ ×2



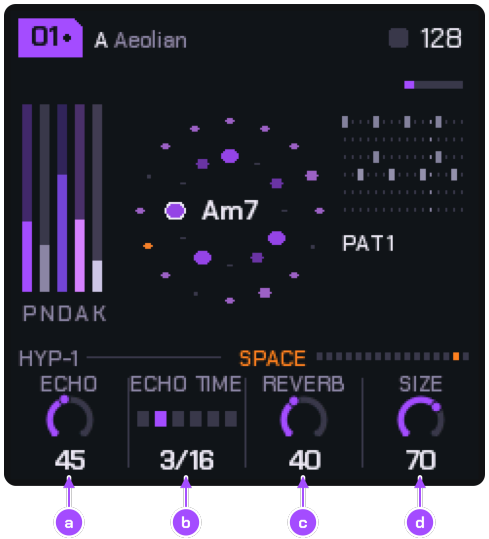

- **a** **K>Drone.** Each kick opens the drone's filter: a rolling bass. Amount: how far and how long.
- **b** **S>Pad.** Each snare or clap flashes the pad open and swells it.
- **c** **H>Acid.** A hat on an acid note accents it.
- **d** **P>Clock.** Off or Clock: the acid steps only on the perc lane's hits. SEL + SEQ: all links on / off.

## Kit REACH IT: SEQ ×3



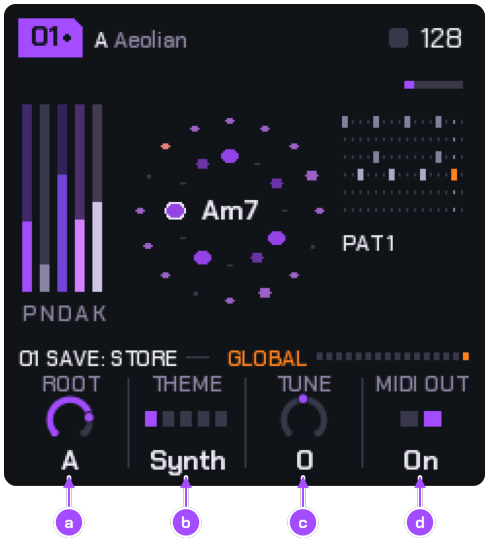

- **a** **Kick.** Hypno, House, Techno, Acid or Break (the Break kick chops and crushes the drums live).
- **b** **Snare.** The snare's sound, and the clap's with it.
- **c** **Hats.** Closed and open hats.
- **d** **Perc.** The perc. SEL + ALGORITHM: all four at once.

## Space REACH IT: SELECT




- **a** **Echo.** Send to the tape echo.
- **b** **Echo time.** 1/8, 3/16, 1/4, dotted 1/4, 1/2, 1 bar.
- **c** **Reverb.** Send to the plate.
- **d** **Size.** How long it rings, and how often the echo returns.

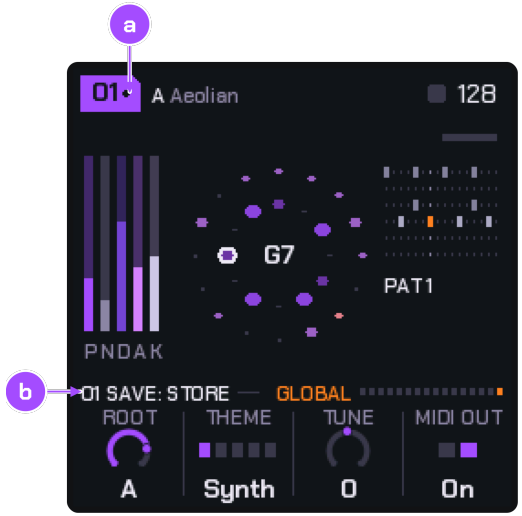
## Global REACH IT: GLO

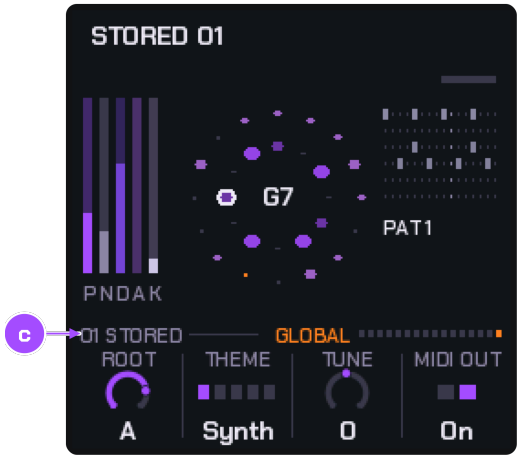



- **a** **Root.** The key's root note.
- **b** **Theme.** Synth, Acid, Ice, Ember, Phosphor.
- **c** **Tune.** ±50 cents.
- **d** **MIDI out.** MIDI out over USB on or off.

# Pieces and storing

PRESETS (3) CHOOSES · GLO (17) THEN SAVE (19) STORES · HOLD SAVE 1.5 S: FACTORY




| # | Character | BPM | Key |
|---|---|---|---|
| **01** | Hypnotic four on the floor | 128 | A Aeolian |
| **02** | Deep and dubby: rims, shakers | 124 | D Dorian |
| **03** | Pounding: rumble kick, claps | 132 | E Phrygian |
| **04** | Trance: offbeat opens, rolls | 138 | G Aeolian |
| **05** | Acid electro: cowbells, breaks | 130 | A Dorian |
| **06** | Dub techno: sparse, swung | 120 | F Pent minor |
| **07** | Minimal glitch: threes on fours | 134 | B Pent major |
| **08** | Rolling, tom-like percussion | 126 | Bb Mixolydian |
| **09** | Warehouse: hard, rides, rumble | 136 | C Phrygian |
| **10** | Deep field: half time, breaks | 118 | Eb Aeolian |

Each piece has its own settings, its own **six drum patterns** and its own drum sounds. Patterns 1–6 run roughly intro, groove, build, peak, variation, tool.

a **Changed.** A dot in the tag: the piece has changes that aren't stored.

b **SAVE: STORE.** On Global the line above the knobs says so. Press SAVE (19).

c **01 STORED.** Written at once. Playing, the music pauses for a moment (STORING) and carries on where it was. FACTORY if the piece isn't stored.

**What a store holds.** Everything about the piece: every page's values, the scale and mode, the six patterns and the drums' level, tune and decay. Theme, tune and MIDI out are yours across all pieces and are kept on their own.

**Factory.** Hold SAVE on Global for 1.5 s (the screen says HOLD: FACTORY): the piece goes back to how it shipped and leaves the store.

**The pause.** The FM-1 can't make sound while it writes to its memory, so a store made while playing fades the music out, writes, and fades it back in exactly where it was: a fraction of a second. MIDI clock out pauses with it.

# Sounds, and playing by hand

SOUND KNOBS · SEL (13) + ALGORITHM (4) · STORED WITH THE PIECE

| Voice | Where | Sounds |
|---|---|---|
| **Pad** | Pad page, c | Rhodes, Voices, Strings, Glass, Organ, Warm, Stabs (chords on the off-beats; Pad env Release their length), Dark, Dub (dark saw stabs), Trance (supersaw), Poly, Hollow, Pings (bell stabs), Hits (organ stabs), Rave (supersaw stabs), Sine. |
| **Drone** | Drone page, b | Saws, Warm, Grit, Dirty (a distorted sub), Reese (phasing saws), Pulse, FM (the envelope opens it), Organ, Rumble, Sub, Square, Hoover, Fifths (root and fifth), Metal (clangy FM), Sync, Bright. |
| **Acid** | Acid env, d | Saw, Square (the 303's two), Driven, Sync (swept by the envelope), Pluck (FM), PWM, Chip, Sub; arp leads: Trance (three saws), Bleep (a sine), Fifth, Bell, Organ, Laser (pitch falling), Fold, Soft (a triangle). |
| **Kit** | Kit page | Each drum from eight: Hypno, House, Techno, Acid, Break, Minimal, Hard, 808. Mix them: a Techno kick under House hats is fine. |

**Changing a sound.** Turn the voice's Sound knob, or hold SEL and turn ALGORITHM on any of its pages (on the drums' pages that swaps all four drums). The voice carries on: its notes, envelope, LFO and filter stay. The sound's name shows at the left of the line above the knobs.

**Playing a voice by hand.** The keys set the scale. On a voice's page, hold SEL and press ENV to take that voice off the algorithm: it goes quiet (BY HAND on the line above the knobs), and on its pages the keys play it over the song. The pad: up to four notes. The drone: the last key held, two octaves down. The acid: the last key held, sliding to a second key as a 303 does. SEL + ENV again gives it back. It follows the OCT buttons, and works while stopped.

**Older pieces.** A piece stored with a style before 1.0.27 comes back with that style's sounds and kit.



Pad: Glass



Acid: Sync



Drone: Reese, by hand



Kit: Techno

# The step sequencer

HOLD SEQ (21) · TAP SEQ TO LEAVE




a **Level.** The selected drum's volume.

b **Tune.** Its pitch, ±1 octave.

c **Decay.** How long it rings.

d **Pattern.** Which of the 6 patterns you're editing (even before it plays).

e **Pattern name.** The pattern on the grid.

f **Selected drum.** The highlighted lane: the one the keys edit. OCT− (10) / OCT+ (11) change it.

g **Lanes.** Kick, snare, clap, closed hat, open hat, perc.

h **A hit.** A filled square is a hit; a dot is a rest.

i **Playhead.** The column playing now.

**Editing.** The 16 white keys (24) are the 16 steps of the selected drum. They light where it hits, and the light running across them is the playhead. Tap a key to turn its hit on or off.

**Shift (hold SEL).** + black keys 1–6: copy this pattern into pattern 1–6. + OCT−: clear this drum's lane. + OCT+: put back the piece's own lane. + white keys 1–6: edit that pattern.

**Keeping it.** Patterns and the drums' sounds belong to the piece. Store the piece (Global, SAVE) and they come back at power-on.

# Effects

FX (12) OPENS IT · HOLD KEYS · HOLD SAVE (19) + TAP: LOCK · FX AGAIN CLOSES



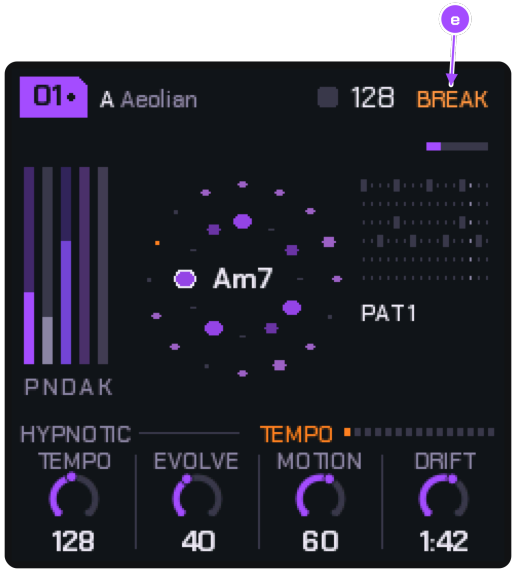

- **a** **Status.** HOLD A KEY: effects play only while their key is held.
- **b** **Key and effect.** Each line is a key and what it does.
- **c** **Held.** Orange: playing now (here G3 and G4 are held).
- **d** **Mute / solo.** M: mutes a track, S: solos it.
- **e** **Lock mode.** SAVE held: each key tapped locks its effect, or unlocks it. Keys held as SAVE goes down lock too.
- **f** **Locked effect.** Purple: on and locked. It stays while you play other effects. Tap its key to turn just it off; closing FX lets them all go.

## Key map

| Keys | Notes | Effect |
|---|---|---|
| White 1–6 | F3 A3 B3 C4 D4 G3 | Beat repeat 1/2, 1/6, 1/8, 1/16, 1/32, 1/4 — starts on the next sixteenth; the shortest held wins. |
| White 7–8 | E4 · F4 | Reverse · Tape stop. |
| White 9–14 | G4 A4 B4 C5 D5 E5 | Mute kick, hats, snare (with clap and perc), drone, acid, pad (with noise). |
| White 15–16 | F5 · G5 | Low-pass closing in · High-pass closing in. |
| Black 1–5 | F#3 Ab3 Bb3 C#4 Eb4 | Gate 1/16, gate 1/8 (a trance gate on the synths), echo throw, bit crush, drum fill. |
| Black 6–11 | F#4 Ab4 Bb4 C#5 Eb5 F#5 | Solo kick, hats, snare, drone, acid, pad — each beside its mute. |

# Performing

SECTIONS, FREEZE, CHORDS




Breakdown: the section shows BREAK (e).




Build: four bars, then the drop.

**Breakdown — SAVE (19), tap.** The breakdown starts on the next bar (the section blinks while it waits) and lasts until you tap again.

**Build — tap SAVE again.** A four-bar build, then the drop.

**Drop — hold SAVE.** Straight to the drop on the next bar. SAVE lights while a breakdown or build runs, and blinks while one waits.

**Freeze — REC (23).** Holds the music on its current chord; press again to let it move.

**Next chord — EDIT (16).** Moves on at the next bar.

**Scale — the keys (24, 25).** Each adds or removes its note; the root stays.

**Patterns — SEL (13) + white keys 1–6.** Straight to pattern 1–6, on the bar. The keys light the one chosen (the one still playing blinks).

**Piece on the bar — SEL + PRESETS (3).** The new piece comes in on the next bar. Plain PRESETS changes at once.

**Links — SEL + SEQ (21).** The drums into the voices, on and off: kick opens the drone, snares flash the pad, hats accent the acid, perc can clock it. Amounts on the Links page.

# Themes, power-on and the simulator

GLO (17) > THEME (6)




Synth




Acid




Ice




Ember




Phosphor

## Power-on

The RedtailWorks card (with its red panda), then HYP-1 (any button skips them); then a piece starts at random: as you stored it, or with small random changes to its factory settings if you never stored it. Unstored changes last until the power goes off.

## What's kept

Every piece you stored (Global, SAVE): written when the music is stopped, never mid-performance. The theme, tune and MIDI out: written two seconds after you change them, even while playing.

## Recording

There's no song save: record the audio (or MIDI out, GLO > d) into your DAW and keep the takes you like.

## Simulator: mouse and touch

Drag a knob up/down or scroll over it. Click buttons and keys. Shift-click (or right-click) a button to hold it; click again to let go — that's how to hold SEQ, SAVE or HOME. **New start** powers off and on (stored pieces stay, in that browser); **Forget stored** clears them.

## The FM-1 as the controller

Plug it in, press **Connect MIDI** on the emulator, then SEL + GLO on the FM-1: it goes quiet and its whole panel plays the emulator, lights and all (SEL + GLO again: back). Stock firmware or any controller: the keys play at once; **MIDI learn** maps the rest.

## Simulator: keyboard

| | |
|---|---|
| **Space** | PLAY |
| **Enter** | REC |
| **Esc** | HOME |
| **V** | ENV |
| **L** | LFO |
| **P** | ARP |
| **M** | SEQ (hold: sequencer) |
| **O** | GLO |
| **C** | EDIT |
| **`** | FX |
| **B** | SAVE |
| **N** | SEL |
| **Z / X** | OCT– / OCT+ |
| **Up / Down** | SELECT |
| **Q–I** | White keys 1–8 |
| **A–K** | White keys 9–16 |
| **1–0, -** | Black keys 1–11 |